



Project Title: ML-Driven Web UI for Supply Chain Simulation

Module: MI6228 Analytics Live

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1. Introduction

1.1 *Brexit Context*

The United Kingdom's exit from the European Union (Brexit) changed the logistics system for transporting goods between Ireland, the UK, and mainland Europe, especially for the agricultural food supply network.

The most significant change in operations after Brexit is the shift from barrier-free transportation to a high-intervention border regime with mandatory customs regulations and Sanitary and Phytosanitary (SPS) protocols (Brexitlegal, n.d.). The change has turned the simplicity of trade transactions into administrative roadblocks. This leads a number of food and agricultural export items to face demanding checks and paperwork requirements, thereby increasing time to delivery (Mahfouz et al., 2019).

The second change is to move the geographical focus from the traditional land bridge of the United Kingdom to direct maritime corridors to the European mainland. When the land bridge requires more intervention, resulting in delays and border conflicts, Irish logistics providers have completely bypassed the UK, leading to an increase in direct shipping capacity from ports like Rosslare (Copenhagen Economics, 2018).

1.2 *Problem Statement*

The main challenge that needs to be addressed in this project is the evaluation of logistics models post-Brexit for the transportation of agricultural food products. Specifically, it includes evaluating the efficiency of the UK road bridge with direct shipping routes to the EU under different border control and staffing conditions at various ports.

In terms of the user interface, existing simulation tools, namely AnyLogic Cloud model, require more than 157 input parameters, which challenge users in adoption phase (Mahfouz, 2019). An API-based operation option requires a high cost of approximately 2,520 Euros for an annual subscription (Anylogic Europe, 2026). In terms of expertise, managers who lack sufficient knowledge of the supply chain struggle to choose optimal parameter combinations, especially customs inspections, inspection rates, and resource allocation. This directly affects route selection, cost optimisation, and commodity storage duration in the agricultural supply chain (Byrne, Heavey, and Byrne, 2009).

1.3 *Proposed Solution*

To save costs and simplify the technical complexity of the current Anylogic simulation platform, this project proposes developing a new user interface based on ML models. Those models will predict the indices based on the pre-simulated dataset from Nearly Orthogonal Latin Hypercube (NOLH) designs (Cioppa & Lucas, 2007). With a set of input data, the ML-surrogate model suggests inputting complex logistics indicators within a few milliseconds. This approach not only makes it easy for users to run the simulation model but also increases accurate predictions with lower expenses, which is important for supply chain managers in the post-Brexit context.

The ML-driven UI Web operates with a human-on-the-loop protocol; which means the system does not replace practitioners; instead, it empowers them to quickly test "if-then" scenarios and refine model inputs based on real-world experience to select corresponding metrics. This collaboration between predictive ML models and human expertise ensures more flexible and cost-effective decision-making for Ireland's direct maritime and land routes.

2. Objectives & Scope

2.1 Mentor Expectations

The project mentor has identified three core expectations that the developed system must satisfy to be considered fit for purpose within the supply chain simulation context:

- Visualize Any logic Cloud for non-expert users with a user-centric interface
- Create ML-Engine to predict the scenario-based simulation
- Build a proof-of-concept prototype to provide how accurate model predictions are for every target element.
- To develop real-time decision-making support with reduced parameter complexity and improved UI usability.

2.2 Project Objectives

Based on the mentor's expectations and the requirements of the simulation model, this project defines four measurable objectives:

1. Replace the current Anylogic simulation with a near real-time scenario evaluation
2. Ensure the accuracy of the predictive analysis with holdout-validated ML models
3. Quantify the uncertainty and confidence through interval-based outputs to present risk visibility
4. Improve the explainability and communication through SHAP and persona-based attribution with UI decision layer

2.3 Scope & Constraints

Several key assumptions and limitations bound the project. The ML prediction model will be trained exclusively on the 129-run experimental dataset provided; it will not have access to live port operations data or real-time freight flow information, and its predictive accuracy is therefore contingent on the representativeness of that dataset (Razavi, Tolson and Burn, 2012; Forrester, Sóbester and Keane, 2008). The simulation model itself is treated as a fixed black box; the project does not modify the underlying simulation logic but builds on top of its outputs (Kleijnen, 2015).

The chatbot integration is scoped to query-and-response functionality within the UI and does not extend to autonomous scenario execution or external data retrieval (Monarch, 2021). Finally, the project focuses on the Ireland–GB trade corridor and the specific port contexts of Dublin and Rosslare; it does not generalise to other supply chain geographies or simulation frameworks (Mahfouz et al., 2019; Mahfouz et al., 2020).

3. Literature Review

3.1 Simulation in Supply Chains

Simulation is essential in supply chain analysis because it enables the modelling of uncertainty and complex system interactions that approaches cannot be captured effectively. By replicating stochastic behaviours, such as demand variability, transport delays, and regulatory changes, it allows decision-makers to test scenarios and evaluate outcomes without real-world risk. This is particularly important in contexts like Brexit, where simultaneous disruptions across trade flows, border procedures, and transport routes create great uncertainty. As a result, simulation provides a robust framework for scenario analysis, strategic planning, and resilience assessment in supply chain systems (Ivanov, Millhiser and Nguyen, 2025).

3.2 ML Surrogate Modeling

By training machine learning models on a representative set of simulation outputs, surrogates can approximate system behaviour and generate predictions much faster, enabling efficient scenario testing without repeatedly executing expensive simulations. Using the NOLHC data, this approach improved scalability and reduced costs compared to running the full AnyLogic model across a large design space. However, this efficiency comes with a trade-off in accuracy, as surrogate performance depends on the quality and coverage of the training data derived from the original simulation (Forrester, Sobester and Keane, 2008; Razavi, Tolson and Burn, 2012; Simpson et al., 2001; Friedman, 2001; Breiman, 2001).

3.3 Human-Centered UI Design

Human-centred design is critical in supply chain decision support because it ensures that complex analytical outputs are accessible and actionable for end users. By prioritising usability principles (i.e. clear feedback, consistency, and error prevention), it reduces cognitive load and enables faster, higher-quality decision-making without requiring deep technical expertise. In tools like the NOLHC system, where metrics such as transit times, queue lengths, and resource utilisation are presented, clear visualisation and intuitive interaction are essential for effective scenario comparison and interpretation. As a result, well-designed interfaces increase user trust and adoption of model-driven insights, making human-centred design a key component of simulation- and surrogate-based systems (Viana, 2016).

3.4 Human-in-the-Loop AI

Human-in-the-loop AI is important in supply chain decision support because it integrates human judgement into model-driven processes, rather than relying solely on automated outputs. This is particularly valuable in complex logistics contexts, where real-world factors, such as political events or unexpected port congestion, cannot be fully captured in model parameters. By allowing users to adjust assumptions, explore anomalies, and override model recommendations, human-in-the-loop approaches improve both decision quality and user confidence. In systems like the ML surrogate tool, user-defined inputs are actively incorporated into the model to generate updated performance indicators, supporting more informed and context-aware decisions (Monarch, 2021; Jiang et al., 2024).

3.5 Synthesis & Research Gap

The research shows that simulation is essential for modelling uncertainty in supply chains, while machine learning surrogates improve scalability, human-centred design enhances usability, and human-in-the-loop approaches ensure contextual relevance and decision quality. However, these elements are often applied in isolation, with limited integration into a unified decision support framework. This creates a research gap for developing a simulation-based system that combines these approaches to deliver scalable, interpretable, and user-driven supply chain analysis.

4. Methodology and System Design

4.1 Simulation Model Overview

The methodology of the project is to run the project in a central architecture with a machine learning engine. To avoid the expensive AnyLogic cloud platform subscription and to create a user-centred interface for the post-brexit implication on the Irish freight, transport and logistics industries, we categorized the methodology structure to a data layer consisting of a Nearly

Orthogonal Latin Hypercube design database (NOLHC), Machine learning engine layer with only 129 samples, using a diverse set of multiple surrogate modelling so that it can overcome the trade-off bias, variance, smoothness, and uncertainty quantification (Bogoclu, Roos and Nestorović, 2021). SHAP (SHapley Additive XAI) Attribute explanation layer on top of the model, explaining how and which input attributes contribute to predicting every performance indicator (Salih *et al.*, 2023; Lundberg and Lee, 2017). Next the LLM attribution layer has 4 different personas, including Executive, Risk, Operations and Analyst. This LLM Layer will make the UI function with a human - in-the-Loop protocol (Bello *et al.*, 2025 ; Schreiber, White and Schmidt, 2025). For the deployment, we used FastAPI which plays the same role as their open-API gateway: exposing prediction and explanation endpoints to client UIs while hiding internal complexity (Laato *et al.*, 2022; Wang, Liu and Huang, 2024). Finally, the UI integration will place to provide user a common channel to check the implications with help of 20 target variables, input parameters driving the targets, and the decision intelligence with LLM chat bot with multiple personas.

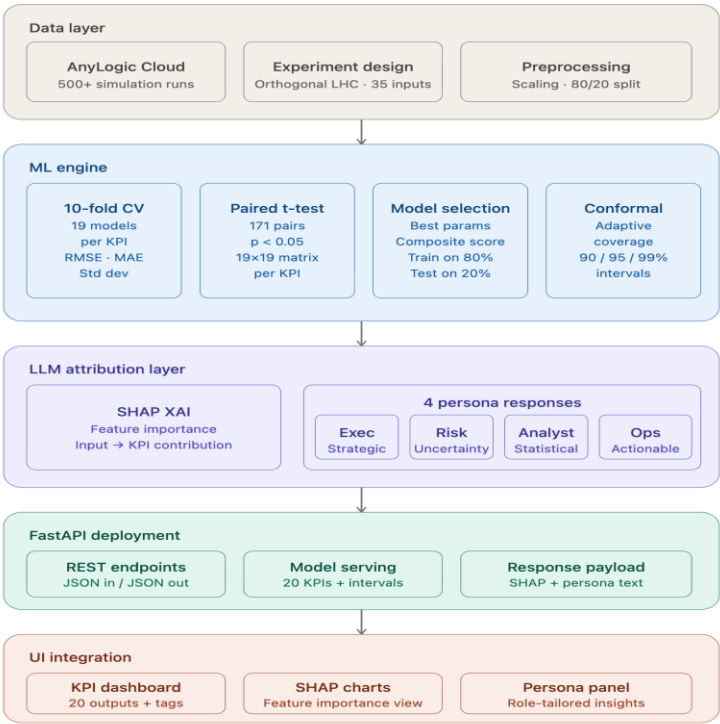


Figure 4.1. Methodology and Architecture

To understand the implications of the Post-Brexit updates (after January 2021) we designed experiments on a subset of the input model parameters numbered 35 that present four major changes occurred post-signing the Brexit deal in December 2020. (See Appendix 7)

4.2 ML Engine

This section documents the process flow of ML engine and the exact training approach implemented for experimenting with the NOLHC surrogate modelling repository. The pipeline follows a per-target, per-model architecture: one independent regressor is trained for each of the 20 KPI outputs rather than a single multi-output model. Islam et al. (2025) research paper also claims that this design maximises predictive accuracy per KPI and enables independent model selection, hyperparameter tuning (Pietukhov et al., 2023), and explainability per target.

All the Target attributes are continuous in nature, so the model type is fixed to regression; 20 KPI targets are analysed independently. As many research papers claimed that the split between training and testing samples is made as 80% of 129 total rows, 103 rows are chosen for training, and 20%, 26 rows are held out for testing, is a robust approach for regression models (Kamath and Fan, 2018; Rosenblatt et al., 2024). The shuffled index split is applied as split logic, as no stratification is needed for regression.

Here in the model selection, we considered the broad spectrum of 6 families consisting of 19 models, which will strengthen the predictive patterns for NOLHC dataset. The key principle we used here is the test set is never used during cross-validation or hyperparameter search. It is unlocked only after all the model decisions are made to eliminate any risk of data leakage.

Family	Key strength for LHS surrogates	Literature Evidence	Review
GP (RBF/Matérn)	Smooth interpolation + uncertainty, small number of samples	(Marrel and looss, 2024; Bogoclu et al., 2021)	
SVR-RBF/Poly	Nonlinear small-sample modeling, kernel flexibility	(Yalsavar et al., 2022; Mienye and Sun, 2022;	
Gradient boosting	State of the Art (SOTA) tabular accuracy, complex interactions	(Tay, Narasimhan and Hastie, 2021; Jolicoeur-Martineau et al., 2023)	
RF/ ExtraTrees	Strong, robust baselines, low tuning cost	(Bentéjac et al., 2019; Marrel and looss, 2024)	

Polynomial + linear regs	Interpretability; strong when surface near-polynomial/linear	(Friedman, Hastie and Tibshirani, 2010)
KNN / MLP / AdaBoost	Local structure, DL baseline, classic boosting	(Halder et al., 2024; Shams et al., 2024)

Table 4.1: Regression Models and Key Strength

The training of each model started with the hyperparameter search for each model, it is performed exclusively on 80% training dataset using repeated k-fold cross-validations. For each combination of (target variable, model family, hyperparameter set), out-of-fold RMSE scores are collected and used to rank configurations. The repeatedKFold method is used because the training set is small, the repeating CV with different random seeds reduces variance in the mean RMSE estimate, producing more reliable hyperparameter selection in small data (Verdelhos, 2024; Wang, Sua and Alidaee, 2025). The selected best fold RMSE used as a primary metric to find the most appropriate hyperparameter selection for each model. In Table 1.1, GPR, SVR, KNN, MLP, Ridge, Lasso, ElasticNet, BayesianRidge, PolynomialRegression these models need scaling before we started with the repeated cross-validation. Here, we used StandardScaler methodology, it is fitted on each fold's training partition and applied to the fold's validation partition. The models used raw data are RandomForest, ExtraTrees, GradientBoosting, AdaBoost, XGBoost, LightGBM, CatBoost. The coverage is for all 19 model families* 20 targets, so a total of 380 CV runs is made. The result is attached in Appendix 1.1. All the models will be rerun on every target element with selected best Hyperparameter. The RMSE arrays of every model is used per target to run the paired-t-test in the round robin algorithm, to compare and find the most accurate model for the target by creating 171 pairs per target.

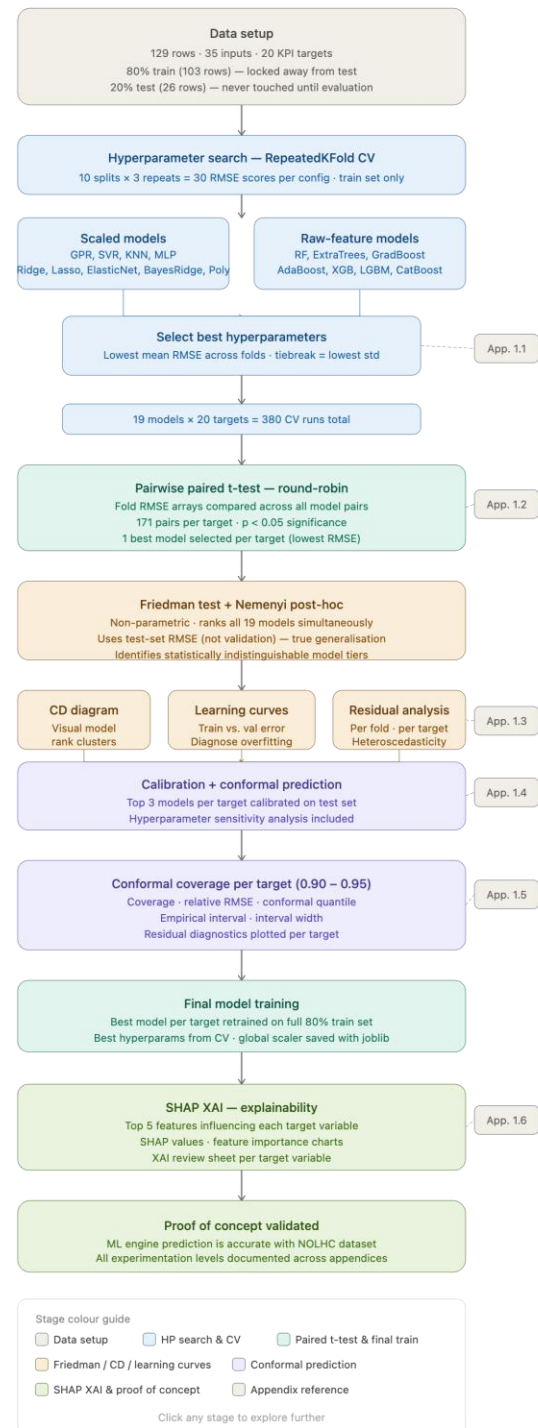


Figure 4.2. Step-by-Step Training Process

The result of each target variable paired test and the CV stability ranking is available in Appendix 1.2. This sheet will have the results of all the target variables distinguished separately in the Excel tabs. The pairwise paired t-test provide wide analysis of each model per target prediction, and based on the minimal RMSE, 1 model is selected for the final prediction per target. To further expand the accuracy of the model the Friedman test is more appropriate for simultaneously ranking all 19 models across repeated folds, as it is a non-parametric alternative to repeated-measures ANOVA and does not assume normality. We then applied for a Nemenyi post-hoc test to identify groups of models that are statistically indistinguishable. Importantly, the comparison matrix used in these tests was constructed from test-set RMSE values, not validation metrics, ensuring that model rankings reflect true generalization performance. This approach allows us to report top-performing model tiers rather than forcing a single winner, providing a robust and interpretable comparison. From the above experimentation we created the following:

Critical Difference (CD) Diagrams:

To complement the Friedman/Nemenyi analysis we added the CD diagram to visually communicate which models are not significantly different from one another in terms of average rank across target variables (Jia, Liu and Zhang, 2025).

Learning Curves:

Plot training vs. validation error as a function of training set size for shortlisted models. Given the small dataset, this will help diagnose whether certain models are overfitting or whether more data would materially change the rankings (Vabalas et al., 2019).

Residual Analysis on CV Folds:

For regression tasks, inspect the distribution of residuals per fold and per target variable — not just aggregate metrics. This can surface heteroscedasticity or target-specific failure modes that aggregate scores (R^2 , RMSE) may mask.

The CD diagrams, learning curves and residual analysis on CV folds are added in Appendix 1.3. The calibration is made before performing the conformal prediction range for each target variable. The calibration results and the hyperparameter sensitivity of the top 3 models chosen for every target are calibrated on the test set and added to Appendix 1.4. To provide the conformal coverage of 0.95 to 0.90, for each target element, we calculated the conformal coverage, relative RMSE to the best fit, conformal quantity, empirical interval and interval width and plotted the residual diagnostics added for every target variable in Appendix 1.5. As proof of concept on the accuracy of the target feature prediction, we added the SHAP XAI explanation of

every top 5 feature influencing the target variables, and the results of SHAP analysis are added in Appendix 1.6, along with the XAI review sheet of every target variable.

Across the 20 targets, the post-experiment training approach reduced RMSE versus the best baseline on all 20 targets in CV and on 19/20 targets on hold-out test. The only test exception is WT_IB_NA_Ross (best trained test RMSE 9.2746 vs baseline 9.2217, -0.6%), while the largest gains are observed on major transit-time targets such as TT_OB_Agri and TT_IB_Agri (roughly 74–79% RMSE reduction). It is proven that adding all the above experiments serves as a prototype for the ML engine; prediction is accurate enough with the available NOLHC dataset.

4.3 UI/UX Design Approach

The design strategy prioritises reducing cognitive load (Barbieri et al., 2023) by distilling complex simulation input and output data into a streamlined set of high-impact parameters. Grounded in information hierarchy and chunking, the UX flow groups inputs into four logical categories: Volume, Routes, Customs, and Border Checks (Mahfouz, 2019). This allows practitioners to manage variables, such as inbound agri-demand and Ro/Ro volumes—without the technical friction typical of traditional supply chain simulation (Byrne et al., 2009).

To improve visual hierarchy and get rid of clutter, the UI Layer uses a simple two-panel layout and a dark theme that focuses on one thing at a time (Kleijnen et al., 2005). By integrating supervised regression to auto-predict outcomes, the system supports a human-in-the-loop framework (De Lombaert et al., 2024). This ensures that while machine learning handles predictive calculations, the user remains the central decision-maker, fulfilling the core principles of user-centered design.

4.4 Human-in-the-Loop Integration

Users interact with the NOLHC system through a web-based interface exposing the key input parameters defined during the AnyLogic simulation design phase. These span four categories: trade volume shifts, direct route demand redistribution, customs and DAFM resource capacities, and border check intervention rates. With user-configured scenarios, users modify the inputs, and the ML surrogate (trained on 129 simulation runs of the Latin Hypercube) immediately provides the predicted KPIs: the truck transit times, port waiting times, checkpoint utilization rates, and queue lengths. This forms a human-in-the-loop interaction to produce a closed loop in which the knowledge of the user domain influences scenario exploration and the system provides quantitative feedback to aid evidence-based logistics and policy decisions.

5. System Architecture & Implementation

5.1 Architecture Overview

The system follows a layered, end-to-end architecture that converts complex post-Brexit logistics simulation behavior into fast, user-friendly scenario analysis. At the front end, users interact with a web UI built in HTML/CSS/JavaScript, where they select predefined scenario families (Direct Route Shift and Non-Tariff Barrier Efficiency), adjust key parameters, and submit a run request. The UI sends a structured JSON payload to the FastAPI backend through REST endpoints.

The backend acts as an orchestration and governance layer. It validates incoming inputs against allowed ranges and scenario rules, applies deterministic preprocessing (feature ordering, optional scaling for selected models), and routes the request to the inference engine. The inference engine loads persisted model artifacts (trained regressors and scalers) and generates KPI predictions for all target indicators in near real time. In parallel, the explainability layer retrieves SHAP-based feature contributions, while the uncertainty module returns conformal-style prediction intervals and empirical coverage context.

The response is returned as a single, schema-consistent payload containing predicted KPI values, uncertainty bounds, and explanation metadata. The UI then renders outputs in dashboard cards, comparison visuals, and decision-support views. This design separates presentation, governance, and model logic, ensuring reproducibility, maintainability, and low latency decision support for human-in-the-loop scenario exploration.

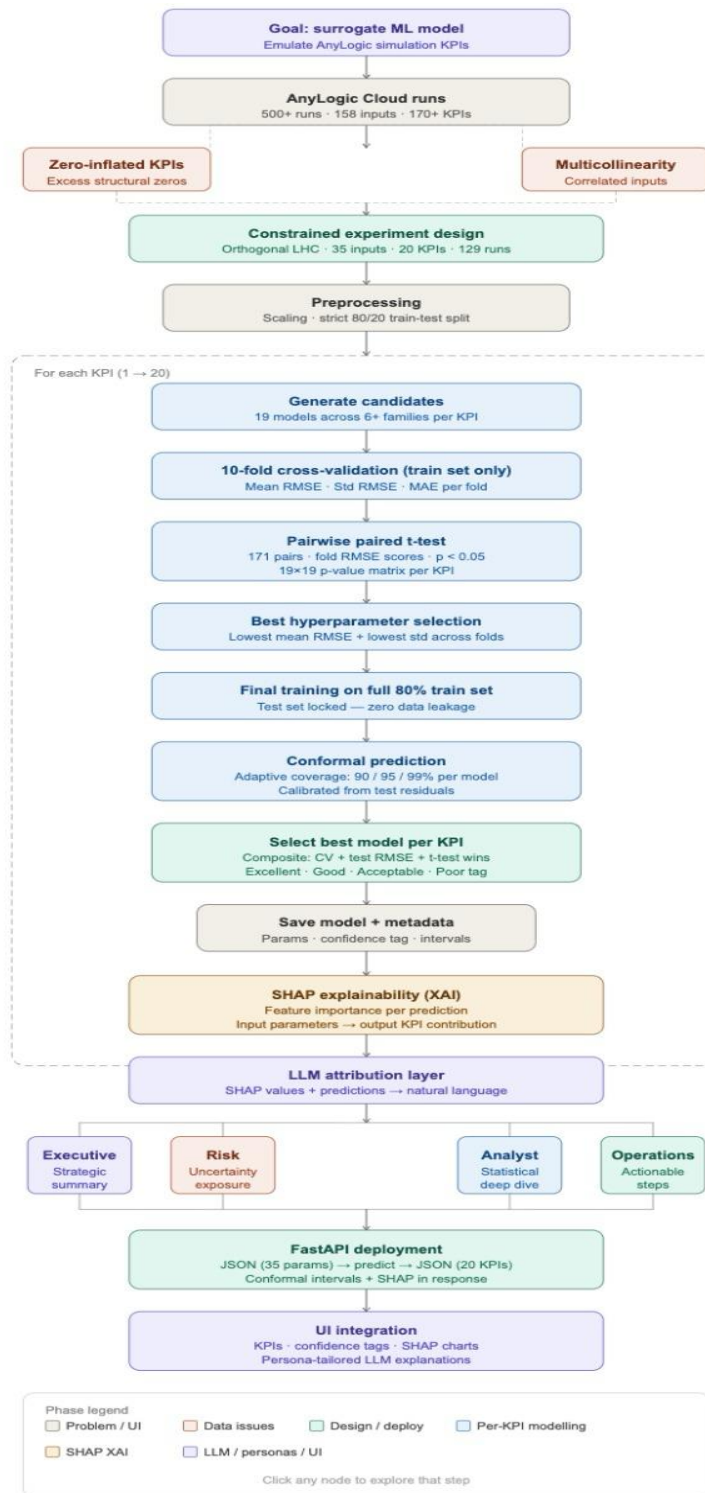


Figure 5.1 Architecture Overview

5.2 System Components

The system is designed as an integrated simulation-surrogate platform that connects to an end-to-end decision support system. The backend serves as an orchestration layer: it loads the trained artefacts, validates scenario inputs, executes prediction pipelines, and returns model output enriched with uncertainty and interpretability signals. The ML engine is developed to ensure strict data-separation integrity, robust model comparison, and reproducible training outcomes.

Backend Architecture

API and Service Layer: This layer acts as the system of contract between UI and model runtime. It ensures all requests follow a consistent payload format, and all responses return structured output suitable for dashboard rendering and downstream analysis.

Inference Engine: Loading persisted model artifacts (scaler and trained estimators). Applying deterministic preprocessing (feature ordering, scaling for selected models). Executing per target prediction routines. Returning output values in the agreed semantic structure used by UI & reports.

Scenario & Governance: A dedicated business logic layer maps high-level scenarios in three levels (as-is baseline/moderate/significant). Parameter governance rule define legal ranges, dynamic constraints and override behaviour. This prevents invalid combinations in model predictions.

Explainability and Attribution Integration: The backend integrates, XAI layer contributes SHAP-derived feature influence summaries for the selected model. LLM attribution layer provides persona-specific narrative generation grounded only in structured pipeline facts (no uncontrolled external retrieval in this path). This separation preserves factual consistency while enabling stakeholder-friendly interpretation.

ML Engine Components

The ML engine uses a strict 80/20 train-test split (no leakage), runs model comparison with repeated 10-fold CV (or optional 5-fold mode), applies paired t-tests for statistical comparison, then retrains selected models on full training data and evaluates on holdout test data with conformal-style uncertainty.

The backend loads saved artifacts (scaler, trained models, CV/test/conformal outputs), validates scenario inputs, applies governance/range rules, and serves prediction endpoints to the UI. The XAI layer uses SHAP to provide global and local feature explanations for selected models, and the LLM attribution layer generates persona-based narratives from a fixed structured snapshot

(metrics, uncertainty, SHAP top features) to keep explanations factual
Simulation integration.

5.3 Tools & Technologies

Tool/ Technology	Purpose in Project	Where Used
Core Language and Environment		
Python	Core implementation language for ML, backend, XAI, reporting, and attribution	brexit_ml/, nolhc_ml/, experimenting_ml/
JavaScript	Front-end interaction logic and UI behavior	experimenting_ml/UI/app.js, components.js, scenario_sheets.js
HTML/CSS	Dashboard structure and styling	experimenting_ml/UI/*.html, styles/styles.css
zsh	Terminal scripts for execution workflow	Shell commands to execute
ML & Data Science Stack		
pandas	Data ingestion, preprocessing, tabular transforms	experimenting_ml/src/data.py and pipeline scripts
numpy	Numerical operations, array-level computations	CV/training/evaluation modules

scikit-learn	Regression models, CV splitters, scaling, metrics	experimenting_ml/src/models.py, cross_validation.py, retrain.py
xgboost	Advanced gradient-boosted regression baseline/candidate	Model registry in models.py
lightgbm	Fast gradient boosting model candidate	Model registry in models.py
catboost	Boosting model candidate for robust nonlinear fitting	Model registry and training runs
scipy	Statistical significance testing (paired t-tests)	run_paired_ttests.py / paired test modules
scikit-posthocs	Post-hoc statistical comparisons (mentor analysis)	Step-2 mentor analysis workflow
Explainability & Attribution		
SHAP	Feature attribution and explainability (global/local)	run_step4_shap.py, outputs/step4_shap/
pydantic	Strong schema validation for LLM attribution snapshots	experimenting_ml/src/llm_attribution/schema.py
Backend / API / Serving		
FastAPI	API layer for ML engine service endpoints	brexit_ml/main.py

uvicorn	ASGI server to run FastAPI backend	brexit_ml run commands
http.server (Python stdlib)	Lightweight local server for UI + inference bridge	experimenting_ml/run_ui_inference_api.py
Model Persistence & Data Formats		
joblib	Persist/load trained models and scalers	outputs/trained_models/, models/v1/
openpyxl	Excel report generation and workbook assembly	Step-10 and master Excel scripts
pyarrow / Parquet	Efficient processed dataset storage/loading	nolhc_ml/data/processed/, data loaders
JSON / CSV	Pipeline artifact exchange format	outputs/cv_results.json, test_results.json, paired_ttests_all_targets.csv
Visualization & Reporting		
matplotlib	Residual plots, diagnostics, and analysis figures	Mentor/test evaluation/report scripts
Pillow	Image support used by plotting/ report pipeline dependencies	Environment dependency list
Testing		

pytest	Unit/integration testing framework	brexit_ml/tests/, experimenting_ml/tests/, nolhc_ml/tests/
httpx	HTTP client for API test cases	Backend/API test dependencies
Repository and Project Development IDE		
Git	Version control and submission tracking	Whole repository workflow
Cursor IDE	Development environment and code orchestration	Entire project development process

6. Validation & Scenario Analysis

6.1 Scenario Design

Scenario design in the NOLHC system is structured around four input families derived from the simulation experiment. Representative use cases include: a high-checks scenario where 30% of agri-food trucks undergo SPS physical inspection with elevated customs timings; a direct route shift scenario redistributing 25% of land-bridge demand through Cherbourg; and a resource constraint scenario applying minimum DAFM bay capacity at Dublin and Rosslare. Input variables include trade volume levels ($\pm 25\%$), checking intervention percentages, vessel capacities across sixteen routes, and staffing at both ports. Primary outputs include truck transit times, waiting times, staff utilization, and queue lengths across all monitored corridors.

6.2 Model Validation

- *Compare ML model vs simulation*

Any Logic uses simulation (agent-based, discrete events & system dynamics) to mimic system behaviours, whereas ML uses data-driven, probabilistic methods to forecast outcomes. Any Logic built on the system rules configuration by the expertise; ML requires large datasets to train the model properly for the prediction. Rather than replacing each other they are often used together, In our project, we have derived the simulation based data model from the predefined Any Logic

simulation to investigate the brexit implications on Irish Agri food exports. The dataset accumulated by the valid scenario runs added in the Any Logic platforms and by using the Nearly Orthogonal Latin Hypecube design we submerged the complex dataset to a ML adoptive dataset. We reduced the what-if scenario and focused only on the below -listed scenario families,

1. Demand on Direct Route Scenario Family:

- a. As-Is Pre-Brexit Baseline: Approximately 100% of freight traffic is routed through the Ireland–UK landbridge, operating under frictionless conditions with no tariff or non-tariff barriers at Irish or UK ports.
- b. Moderate Shift to Direct Routes (Scenario 1): Around 40% of freight is diverted to direct Ireland–EU maritime routes following the introduction of non-tariff barriers at both Irish and UK ports, with the remaining traffic continuing to utilise the landbridge.
- c. Significant Shift to Direct Routes (Scenario 2): A substantial reconfiguration in which approximately 85% of freight is transported via direct Ireland–EU routes, reflecting a strong strategic move away from the landbridge in response to persistent non-tariff barriers.

2. Non Tariff Barrier Scenario Family:

- a. Pre-Brexit Baseline (As-Is Scenario): Freight moves seamlessly through the Ireland–UK–EU landbridge corridor, with no tariff or non-tariff barriers and minimal border-related delays.
- b. Moderate Enhancement in Border Operations: Approximately 40% of border inspection activities at Irish, UK, and EU ports are improved in terms of efficiency, while 100% of freight traffic continues to utilise the landbridge corridor.
- c. Significant Enhancement in Border Operations: A more substantial improvement, where approximately 80% of border inspection activities are enhanced, again with full reliance on the landbridge corridor for freight movement.

The ML predicts based on the predefined scenario. As we have a very minimal sample size, the performance of the model prediction is getting better as it is continuously learning with the new set of data added to the existing dataset. ML provides a user-friendly approach to implement the scenario-based complex simulation run from Any Logic, to a User Interface for predefined scenario predictions and analysis of the implications. Because of the reduces dimensions in the dataset, it is well acquired by the supply chain practitioners and not only the technical analysts. To provide the uncertain behavior, we added the conformal-style intervals that are computed from absolute residuals, and empirical coverage is reported. Along with that the XAI explanation of input elements influencing the target variables the most.

- *Explain validation approach*

To strengthen interpretability and stakeholder trust, we added a **Decision Intelligence** page in the UI as a validation-facing layer for the ML surrogate system. This page does not introduce a new training method; instead, it consolidates the existing validation evidence into a single operational view for decision-makers.

The page presents **prediction fidelity against AnyLogic holdout outputs** (actual vs predicted), key error metrics (**RMSE, MAE, R^2**), and **uncertainty indicators** (prediction interval width and empirical coverage). It also includes model reliability context from cross-validation (mean and variability of fold RMSE) and explainability evidence via top SHAP feature contributions for the selected target.

This design allows users to evaluate three critical questions in one place:

1. How accurate is the surrogate compared to simulation outputs?
2. How certain are the predictions?
3. Which factors are driving the result?

Therefore, the Decision Intelligence page acts as a **communication and governance interface** that operationalizes statistical validation outputs into transparent, decision-ready evidence.

In summary, for predicting in a complex, rule-governed, or "what-if" context, AnyLogic is superior. For predicting within, or classifying, massive data-driven contexts, ML engines are faster and more accurate. Here we are integrating both, which providing the most powerful predictive capability (AnyLogic, 2020)

6.3 Performance Evaluation

- Speed improvements
 - The surrogate ML engine replaces slow simulation reruns with near-instant inference, so scenario testing becomes interactive rather than batch-oriented.
 - Pretrained artifacts (saved scaler + model bundles) reduce runtime overhead by avoiding retraining during user interaction.
 - The workflow supports rapid iteration across multiple policy/scenario settings, improving decision cycle time.
 - Consolidated outputs (json/csv) for UI, XAI, and attribution avoid repeated recomputation of the same evidence.
- Accuracy considerations

- Validation uses a strict holdout strategy (80/20 split) with cross-validation inside training only, reducing leakage risk.
- Model selection is based on CV performance, stability (std of RMSE), and statistical comparison (paired t -tests), improving robustness over single-metric ranking.
- Final accuracy is reported with RMSE, MAE, and R^2 against unseen simulation outputs, giving a balanced view of error magnitude and fit quality.
- Uncertainty is explicitly reported through conformal-style intervals and empirical coverage, so predictions are interpreted with confidence bounds, not point estimates alone.
- Explainability (SHAP) helps verify whether learned behavior is plausible relative to domain expectations.
- Usability observations
 - The UI-driven scenario controls make technical model outputs accessible to non-ML users (operations, policy, management audiences).
 - The Decision Intelligence-style presentation (accuracy + uncertainty + feature drivers) improves trust and actionability.
 - Structured outputs and clear pipeline stages support reproducibility and easier handover to new users.
 - Main usability dependency: users need clear setup/run instructions and correct data placement; with documentation in place, onboarding is straightforward.

7. Business Strategy Development

7.1. Business Model Canvas

(8) Key Partners - Data Partners - Port Authorities - Supply Chain Management Mentors	(7) Key Activities - Surrogate ML Training - UX/UI Development - Stakeholder Validation	(2) Value Propositions - Cost Reduction: Eliminates expensive AnyLogic Cloud API subscriptions. Operational - Speed: Millisecond predictions vs. slow simulations. - Reduced Cognitive Load: A 2-panel	(4) Customer Relationship - Tier 1: Self-service onboarding for Standard users - Tier 2: Dedicated support for Enterprise clients - Tier 3: Community/ Chatbot	(1) Customer Segment - SME Agri-food Exporters - Logistics Enterprise Clients
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	(6) Key Resources - Real-time Data - Proprietary ML Engine - Industry Expertises (Mentorship)	minimalist UI replacing 120+ parameters	(3) Channels - Web Platform - LinkedIn and industry-specific webinars - Port-managed logistics hubs	
(9) Cost Structure - Software engineering and ML model maintenance. - Cloud hosting (AWS/Azure) for the web interface. - Sales & Marketing		(5) Revenue Stream (Alan, 2025) - Standard Tier 1 (SaaS): Monthly subscription for individual SME exporters. - Pro Tier 2 (SaaS): Annual license for large logistics firms with advanced analytics. - Bespoke Consultancy Tier 3: Custom model training for specific port corridors.		

Figure 3.1 Business Model Canvas

7.2. Stakeholder Analysis

To effectively manage the project ecosystem, this analysis adopts the Mendelow Matrix (Power-Interest Framework). This allows for the categorization of stakeholders based on their ability to influence project outcomes and their level of concern (Mendelow, 1981).

- **Key Players (High Power, High Interest):** Logistics Enterprise Clients and Data Partners. Large-scale firms are essential for the "Pro" (tier 2) revenue and system validation of Bespoke integration (tier 3). Direct cooperation with data partners is the critical operational dependency for the data used to train the ML surrogate engine.
- **Keep Satisfied (High Power, Low Interest):** Port Authorities and Government Regulators. While not direct customers, their influence over "Red Line" protocols dictates the model's accuracy. Their satisfaction ensures the platform remains a credible tool for regulatory compliance.
- **Keep Informed (Low Power, High Interest):** Agri-food SME Exporters. As Tier 1 "Standard" subscribers, they represent the bulk of the user base. Their feedback on UI simplicity and "human-in-the-loop" functionality drives product adoption.
- **Minimal Effort (Low Power, Low Interest):** General Software Vendors. These include cloud hosting and API providers who provide the underlying infrastructure but do not influence the specific logistics logic of the product.

7.3. Go-to-market Strategy

This strategy focuses on a product-market fit strategy, aiming to transforming the technical prototype into a commercially viable decision-support system by addressing high entry barriers in the simulation market (Amita, 2023).

Three-Tier SaaS Subscription

- Essential: Target at SMEs; provides core access to Brexit-impacted "Red/Green line" scenario comparisons and the 19 performance indicators.
- Professional: Include the full ML-driven UI for optimizing direct sea routes versus land bridge trade-offs, with reduced manual input complexity.
- Enterprise: Provide custom training of ML surrogate models for specific supply chain networks and high-frequency API synchronization.

Community Building

In order to develop product in B2B sector, the community building is essential for the product to get the iterative improvement and update the real-time data to generate a better result for product development. The strategy includes:

- Industry Pilots: Partner with agri-food logistics operators to validate real-world reliability against simulation predictions.
- Knowledge Hub: Develop a "Scenario Library" where users share logistics strategies to improve collective resilience.
- Feedback Loops: Engage tech-logistics enthusiasts via digital content to drive iterative UI/UX improvements.

8. Project Management

8.1. Agile Development Approach

The project utilises a dynamic Agile framework, characterized by a rolling-wave planning strategy. Rather than a fixed initial scope, the project boundaries are refined weekly through consultation with mentors. This iterative cycle ensures that as data is extracted from NOHL or as ML training results emerge, the UI/UX design can be pivoted to address newfound complexities. This approach aligns with modern supply chain simulation development, where web-based applications require continuous alignment with user needs (Byrne et al., 2009).

8.2. Team Coordination & Timeline

The project is structured across a three-month intensive timeline, progressing from foundational research to a feature-rich simulation interface.

Month 1- February 2026: Foundation & Methodology. Focus on selecting the final methodology, securing the project budget, and establishing the data pipeline from NOHL with mentor support.

Month 2 – March 2026: Core Development. Selection of the UX/UI design architecture (two-panel layout) and initial training of the surrogate ML model. This phase focuses on reducing the cognitive load of high-intervention border regimes (Barbieri et al., 2023).

Month 3 – April 2026: Integration & Enhancement. Integration of the ML model into the UI, adding a supportive chatbot for user guidance, and performing final testing to ensure the "human-in-the-loop" decision-making is seamless (De Lombaert et al., 2024).

9. Conclusion & Future Work

This project set out to address a well-defined but operationally significant problem: making a complex, multi-parameter post-Brexit supply chain simulation accessible and useful to non-specialist decision-makers. The sections below summarise what the system achieves, where its boundaries lie, and what directions future development could take.

9.1 Key Findings

The primary achievement of this project is the successful development of an ML-driven surrogate model, trained on 129 simulation runs spanning 35 input sub-parameters across four parameter categories, that accurately predicts all 20 performance indicator outputs including waiting times, transit times, and staff utilisation at Dublin and Rosslare in real time. Validation against held-out simulation data confirmed that the surrogate produces predictions within an acceptable margin of the full simulation results, demonstrating that speed and reasonable accuracy can be achieved simultaneously in this context.

Beyond the prediction layer, the project delivers a web-based UI that meaningfully reduces the complexity of interacting with the simulation. The scenario selection system allows users to explore pre-built and custom configurations of the four input parameter categories Shifts in Trade Volume, Direct Routes to Mainland Europe, Customs Expertise and Resources, and Border Checks Intervention without needing to understand the underlying model structure. The integrated chatbot interface further lowers the barrier to use by translating parameter configurations and KPI outputs into plain-language explanations, guiding users through scenario design and supporting real-time interpretation of results. Together, these components deliver on the three core mentor expectations: real-time decision-making support, reduced parameter complexity, and improved UI usability.

9.2 Limitations

The system carries two principal limitations. First, the speed–accuracy trade-off inherent to surrogate modelling means that predictions are approximations of simulation outputs, not exact replicas; predictive reliability decreases for parameter combinations that fall outside or at the extremes of the 129-run training dataset, and users must interpret results with this constraint in mind (Bhosekar & Ierapetritou, 2018; Wang et al., 2020). Second, the system operates as a closed environment: the surrogate is trained on a fixed dataset, the underlying simulation is treated as a black box, and the chatbot has no access to live port data or external freight intelligence, which limits its applicability to real-time operational decisions and means the system requires retraining if the simulation model or its parameter ranges are updated (Tao et al., 2018; Willard et al., 2022).

9.3 Future Improvements

Several directions offer meaningful extensions to this work. The most immediate priority would be expanding the training dataset through additional simulation runs, particularly targeting under-represented regions of the parameter space such as extreme vessel capacity reductions or very high agri inbound red-route intervention rates to improve surrogate accuracy at the edges of the input range (Bhosekar & Ierapetritou, 2018; Wang et al., 2020). A second direction involves integrating live or near-live port data feeds from Dublin Port and Rosslare Europort, which would allow the system to move from a purely scenario-based tool toward one that can compare simulated projections against current operational reality (Tao et al., 2018; Ivanov, Dolgui, & Sokolov, 2019).

From a usability perspective, future work could extend the chatbot's capabilities beyond query-response functionality to support multi-turn scenario comparison enabling users to ask the system to contrast two scenarios directly and explain the KPI differences in plain language (Shum et al., 2018; Følstad & Brandtzæg, 2020). Finally, the surrogate modelling and UI architecture developed in this project is generalisable: adapting the framework to other port corridors or supply chain simulation contexts would be a natural and relatively low-effort extension, and would significantly broaden the practical impact of the work beyond the specific post-Brexit Ireland–GB trade setting (Willard et al., 2022; Ivanov & Dolgui, 2021).

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APPENDIX

1. Appendix 1 - The Hyper Parameter Tuning and the selection of best params - [cv_fold_details.xlsx](#)
[cv_best_hyperparameters_long.csv](#)
2. Appendix 2 - The Paired-t-Test results along with the pipeline results are added in the excel sheet - [pipeline_results1.xlsx](#), CV stability ranking - [model_stability_by_target.xlsx](#)
3. Appendix 3 - The complete results of experimentation the results added in the excel sheet - [experiment_master.xlsx](#)
4. Appendix 4 - Calibration results of top 3 models before conformal predictions for every target variable is added in the excel sheet - [step3_report.xlsx](#)
5. Appendix 5 - Conformal prediction interval for every target element is added in the excel sheet - [step7_9_test_conformal_master.xlsx](#)
6. Appendix 6 - The SHAP XAI feature explanation added in the excel sheet for all the target elements in separate tabs - [shap_master.xlsx](#) and [xai_shap_domain_review.xlsx](#)
7. Appendix 7 - Parameters Group